Project Report

HR Workflow Designer - Full-Stack Implementation

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Project: Tredence Analytics Internship Case Study

Project Overview

A production-grade HR Workflow Designer built as a full-stack React application. The platform allows HR administrators to visually create, configure, and simulate internal workflows such as employee onboarding, leave approvals, and document verifications.

Architecture & Tech Stack

The application is built using modern web technologies focusing on performance, scalability, and developer experience.

- **Frontend:** React 19 with Vite for ultra-fast development and bundling.
- **Canvas Engine:** React Flow (xyflow) for the interactive workflow builder.
- **State Management:** Zustand for a lightweight, subscription-based global state.
- **API Mocking:** MSW (Mock Service Worker) for realistic backend simulation.
- **Forms & Validation:** React Hook Form integrated with Zod schemas.
- **Styling:** Tailwind CSS with a focus on dark mode and premium aesthetics.

Key Technical Decisions

Zustand over React Context

Zustand provides a flat store structure with zero boilerplate. Unlike React Context, it prevents unnecessary re-renders by allowing components to subscribe to specific slices of state.

Dynamic Automation Parameters

The `AutomatedStepNodeForm` dynamically renders input fields based on the action selected. This data-driven approach allows new automation types to be added via the API without requiring any frontend code changes.

Undo/Redo History

Implemented a custom history stack within the Zustand store, allowing users to revert and redo structural changes seamlessly.

Features Implemented

- Interactive Canvas with Dark Theme, MiniMap, and Controls.
- 5 specialized Node Types (Start, Task, Approval, Automated, End).
- Drag-and-drop creation and intuitive node deletion.
- Real-time graph validation including cycle detection and isolation checks.
- Simulation panel with staggered visual execution of the workflow.
- Workflow Serialization (Export/Import as JSON).
- Auto-layout functionality powered by the Dagre engine.

Conclusion & Future Scope

The current implementation provides a robust foundation for HR workflow management. Future iterations would include persistent database storage, workflow templates, and advanced conditional branching logic for complex business rules.